

**FAST NATIONAL UNIVERSITY OF COMPUTER AND EMERGING SCIENCES
KARACHI CAMPUS**

Fall 2025
Computer Networks (CS 3001)



Assignment No. 2

Date: 03/11/2025

Maximum mark: 30

Submission Guidelines:

- Assignment should be completed neatly in handwritten and submitted in pdf format.
- Write your name, roll number and section on top of the first page of the assignment.
- Any form of plagiarism, including AI-assisted content, in any part of the assignment will lead to zero (0) marks for the entire submission.
- The submission portal will close on 3rd November 2025. **Late submissions will not be accepted under any circumstances.** To avoid any unforeseen issues such as electricity or internet problems, students are advised to submit their assignments at least one day earlier. Email submissions will not be entertained.

Question 1:

[6 * 2 = 12 Marks]

- The transport layer provides logical rather than physical communication between application processes. Explain and illustrate how?
- Provide examples of applications where UDP is preferred over TCP.
- Define the role of the **rdt_rcv()** function in the RDT protocol. How does it differ from the **rdt_send()** function?
- Describe the purpose of the **udt_send()** function in the RDT protocol.
- Describe the concept of Software Defined Networks?
- How can we differentiate Software Defined Network from traditional network?

Question 2:

[3 Marks]

Suppose datagrams are limited to 1,500 bytes (including header) between source Host A and destination Host B. Assuming a 20-byte IP header, how many datagrams would be required to send an MP3 consisting of 48,840 bytes? Explain how you computed your answer.

Question 3:**[10 * 1.5 = 15 Marks]**

Write the values of the following fields of the TCP header, in 'decimal' notation format. The stream, from which the values are to be extracted, is written below in hexadecimal notation.

000Y 00Z0 0000 0001 0000 0000 3FF0 0000 000X FFFF 0100 0000 0010 0000

(1) Source Port number, (2) Destination Port, (3) Sequence Number, (4) Acknowledgement number, (5) TCP header length, (6) ACK bit, (7) FIN bit, (8) SYN bit, (9) Window size, and (10) Checksum.

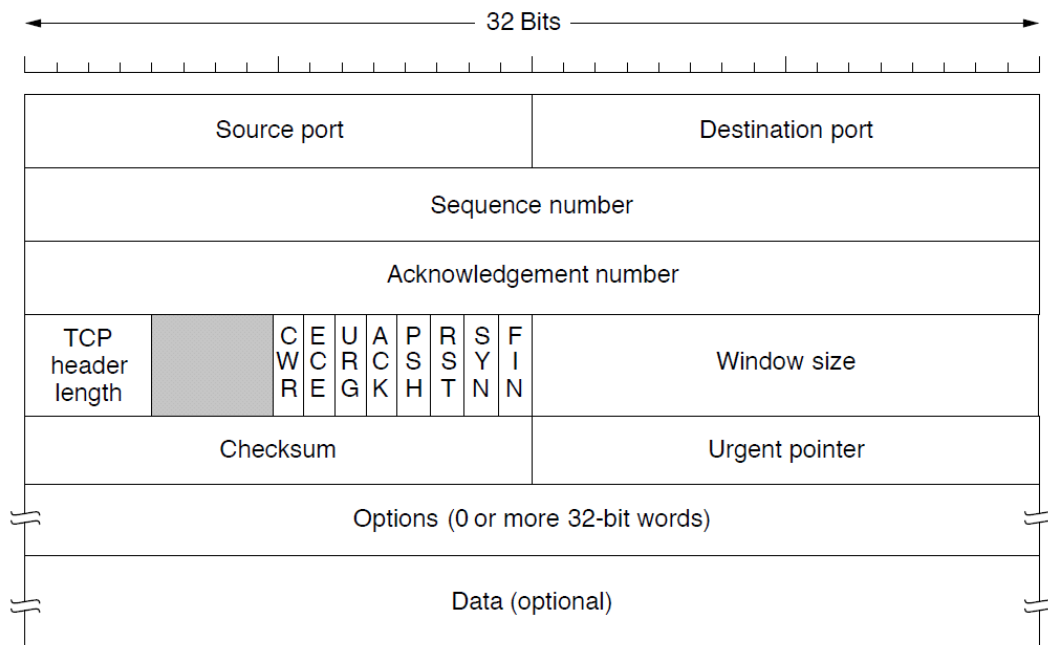
Note!

(a) The given stream is of total 224 bits.

(b) In the given stream, 'Option' field uses 32-bits to show its value.

(c) There is also some contents of the 'data' field, and is shown by the last 32-bit values in the given stream.

(d) X, Y, and Z in the given stream are the parts of student ID number. The most significant digit of the ID will be assigned to variable X, second-most-significant to Y and Z for third-most-significant (i.e. the 2nd last) digit in ID. That is, if the student ID is “20k-1165”, then X=1, Y=1 and Z=6. Every student must have to calculate values of X,Y and Z, as per their IDs to attempt this question.



“THE END”